A Proposed MRTS-Augmented Bayesian Hierarchical Skew-t Model

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1 Goal

The goal of this note is to formulate a computationally efficient alternative to the spatial dependence component in the Morris skew-t hierarchical model. The main idea is to retain the original hierarchical structure for the mean, skew, and scale components, but replace the site-level Matérn correlation model with an MRTS-based low-rank spatial random effect representation. To avoid treating the full $K \times K$ covariance matrix $\mathbf{M}$ as a high-dimensional Bayesian parameter block, we propose to update $\mathbf{M}$ using the closed-form maximum likelihood estimators derived by Tzeng and Huang (2018), and then convert the resulting covariance matrix to a correlation matrix so that the original Morris separation between marginal scale and spatial dependence is preserved.

2 Original Morris Hierarchical Structure

Let $s_1, \ldots, s_n \in D \subset \mathbb{R}^d$ denote the observed spatial locations and let $t = 1, \ldots, T$ index replicates in time. Let $Y_t(s_i)$ denote the response at location s_i and replicate t. For each t, define the response vector

$$\mathbf{Y}_t = \big(Y_t(s_1), \ldots, Y_t(s_n)\big)'.$$

As in Morris et al., we retain the partition-based skew and scale structure. Let P_{tk}, $k = 1, \ldots, K_t^{(p)}$, denote the spatial partitions at time t. For $s \in P_{tk}$, define

$$z_t(s) = z_{tk}, \qquad \sigma_t^2(s) = \sigma_{tk}^2.$$

Let

$$\mathbf{z}_t = \big(z_t(s_1), \ldots, z_t(s_n)\big)', \qquad \mathbf{D}_t = \operatorname{diag}\big(\sigma_t(s_1), \ldots, \sigma_t(s_n)\big).$$

Let $\mathbf{X}_t$ denote the $n \times p$ design matrix for replicate t, and let $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\lambda \in \mathbb{R}$ denote the regression and skewness parameters.

The original Morris model can be written in vector form as

$$\mathbf{Y}_t = \mathbf{X}_t\boldsymbol{\beta} + \lambda \mathbf{D}_t|\mathbf{z}_t| + \mathbf{D}_t\mathbf{v}_t, \qquad t = 1, \ldots, T, \tag{1}$$

where $|\mathbf{z}_t|$ is interpreted componentwise, and $\mathbf{v}_t$ is a zero-mean Gaussian spatial process with a correlation matrix determined by a Matérn family with a nugget-type parameter. In the original implementation, this correlation structure is governed by a small number of hyperparameters such as ρ, ν, and γ, but repeated MCMC updates still require repeated inversions of an $n \times n$ covariance matrix.

3 Proposed MRTS Spatial Layer

3.1 Basis Expansion

Let $f_1(\cdot), \ldots, f_K(\cdot)$ denote prespecified MRTS basis functions, with $K \ll n$ in the intended applications. Define

$$\mathbf{f}(s) = \big(f_1(s), \ldots, f_K(s)\big)',$$

and let the $n \times K$ basis matrix be

$$\mathbf{F} = \begin{bmatrix} \mathbf{f}(s_1)' \\ \vdots \\ \mathbf{f}(s_n)' \end{bmatrix}.$$

Introduce a latent spatial random effect vector

$$\mathbf{u}_t = \mathbf{F}\boldsymbol{\eta}_t + \boldsymbol{\xi}_t, \qquad \boldsymbol{\eta}_t \sim N_K(\mathbf{0}, \mathbf{M}), \qquad \boldsymbol{\xi}_t \sim N_n(\mathbf{0}, \sigma_\xi^2 \mathbf{I}_n),$$

independently over t, where $\mathbf{M}$ is an unknown nonnegative-definite $K \times K$ matrix and $\sigma_\xi^2 \geq 0$ is a fine-scale variance parameter. Conditional on $(\mathbf{M}, \sigma_\xi^2)$,

$$\mathbf{u}_t \sim N_n(\mathbf{0}, \mathbf{G}), \qquad \mathbf{G} = \mathbf{F}\mathbf{M}\mathbf{F}' + \sigma_\xi^2 \mathbf{I}_n.$$

3.2 Standardization to a Correlation Matrix

If $\mathbf{G}$ were used directly in (1), then the marginal variance of the spatial term would be controlled jointly by $\mathbf{G}$ and by $\mathbf{D}_t$, which would blur the original division of labor between the scale process and the spatial dependence process. To preserve the Morris interpretation, we convert $\mathbf{G}$ to a correlation matrix.

Define

$$\mathbf{A} = \mathrm{diag}(\mathbf{G}), \qquad \mathbf{R} = \mathbf{A}^{-1/2}\mathbf{G}\mathbf{A}^{-1/2}.$$

Then $\mathbf{R}$ is positive semidefinite with unit diagonal. We define the standardized latent spatial effect

$$\mathbf{v}_t = \mathbf{A}^{-1/2}\mathbf{u}_t, \qquad \mathbf{v}_t \sim N_n(\mathbf{0}, \mathbf{R}).$$

The proposed hierarchical model is therefore

$$\mathbf{Y}_t = \mathbf{X}_t\boldsymbol{\beta} + \lambda \mathbf{D}_t|\mathbf{z}_t| + \mathbf{D}_t\mathbf{v}_t, \qquad \mathbf{v}_t \sim N_n(\mathbf{0}, \mathbf{R}), \tag{2}$$

with

$$\mathbf{R} = \mathbf{A}^{-1/2}(\mathbf{F}\mathbf{M}\mathbf{F}' + \sigma_\xi^2 \mathbf{I}_n)\mathbf{A}^{-1/2}, \qquad \mathbf{A} = \mathrm{diag}(\mathbf{F}\mathbf{M}\mathbf{F}' + \sigma_\xi^2 \mathbf{I}_n).$$

3.3 Conditional Residual Representation

Given $(\boldsymbol{\beta}, \lambda, \mathbf{z}_t, \mathbf{D}_t)$, define the standardized residual vector

$$\mathbf{r}_t = \mathbf{D}_t^{-1}\big(\mathbf{Y}_t - \mathbf{X}_t\boldsymbol{\beta} - \lambda \mathbf{D}_t|\mathbf{z}_t|\big), \qquad t = 1, \ldots, T. \tag{3}$$

Under the proposed model,

$$\mathbf{r}_t = \mathbf{v}_t \sim N_n(\mathbf{0}, \mathbf{R}).$$

For the purpose of updating the low-rank covariance structure, we also consider the unstandardized low-rank covariance

$$\mathbf{G} = \mathbf{F}\mathbf{M}\mathbf{F}' + \sigma_\xi^2 \mathbf{I}_n,$$

and use the Tzeng–Huang closed-form estimators as a plug-in update for $(\mathbf{M}, \sigma_\xi^2)$ before converting $\mathbf{G}$ to $\mathbf{R}$.

4 Closed-Form Update for $\mathbf{M}$ and σ_ξ^2

4.1 Working Covariance Update

At a given MCMC iteration, treat the residual vectors $\mathbf{r}_1, \ldots, \mathbf{r}_T$ in (3) as the current approximately Gaussian replicates used to update the spatial dependence block. Define the empirical residual covariance matrix

$$\mathbf{S} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{r}_t \mathbf{r}_t'.$$

The spatial random effect model of Tzeng and Huang takes the form

$$\mathbf{z}_t = \mathbf{y}_t + \boldsymbol{\epsilon}_t, \qquad \mathbf{y}_t = \mathbf{F}\boldsymbol{\eta}_t + \boldsymbol{\xi}_t,$$

with

$$\boldsymbol{\eta}_t \sim N_K(\mathbf{0}, \mathbf{M}), \qquad \boldsymbol{\xi}_t \sim N_n(\mathbf{0}, \sigma_\xi^2 \mathbf{I}_n), \qquad \boldsymbol{\epsilon}_t \sim N_n(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I}_n).$$

In the present proposal, there is no separate measurement-error term in the standardized residual update, so the natural choice is

$$\sigma_\epsilon^2 = 0.$$

If a known measurement-error variance is desired in a specific application, it can be inserted without changing the formulas below.

4.2 Theorem 2 Plug-In Estimator

Define

$$\mathbf{H} = (\mathbf{F}'\mathbf{F})^{-1/2} \mathbf{F}'\mathbf{S}\mathbf{F} (\mathbf{F}'\mathbf{F})^{-1/2}.$$

Let the eigendecomposition of $\mathbf{H}$ be

$$\mathbf{H} = \mathbf{P}\, \mathrm{diag}(d_1, \ldots, d_K) \mathbf{P}',$$

where $d_1 \geq \cdots \geq d_K \geq 0$ are the eigenvalues. Let $d_0 = 0$. Define L^* as the largest $L \in \{1, \ldots, K\}$ such that

$$d_L > \max \left\{ \frac{1}{n - L} \left(\mathrm{tr}(\mathbf{S}) - \sum_{k=1}^{L} d_k \right), \sigma_\epsilon^2 \right\},$$

if such an L exists, and let $L^* = 0$ otherwise.

Then the Tzeng–Huang estimators are

$$\widehat{\sigma}_\xi^2 = \max \left\{ \frac{1}{n - L^*} \left(\mathrm{tr}(\mathbf{S}) - \sum_{k=1}^{L^*} d_k \right) - \sigma_\epsilon^2,\ 0 \right\}, \tag{4}$$

and

$$\widehat{\mathbf{M}} = (\mathbf{F}'\mathbf{F})^{-1/2} \mathbf{P}\, \mathrm{diag}(\hat{d}_1, \ldots, \hat{d}_K) \mathbf{P}' (\mathbf{F}'\mathbf{F})^{-1/2}, \tag{5}$$

where

$$\hat{d}_k = \max(d_k - \widehat{\sigma}_\xi^2 - \sigma_\epsilon^2, 0), \qquad k = 1, \ldots, K.$$

The updated low-rank covariance is then

$$\widehat{\mathbf{G}} = \mathbf{F}\widehat{\mathbf{M}}\mathbf{F}' + \widehat{\sigma}_\xi^2 \mathbf{I}_n. \tag{6}$$

To preserve the Morris architecture, define

$$\widehat{\mathbf{A}} = \mathrm{diag}(\widehat{\mathbf{G}}), \qquad \widehat{\mathbf{R}} = \widehat{\mathbf{A}}^{-1/2} \widehat{\mathbf{G}} \widehat{\mathbf{A}}^{-1/2}. \tag{7}$$

4.3 Interpretation

The update in (4)–(7) does *not* treat the $K(K+1)/2$ free elements of $\mathbf{M}$ as a Bayesian parameter block requiring its own MCMC updates. Instead, it profiles the covariance block out by replacing $(\mathbf{M}, \sigma_\xi^2)$ with closed-form estimators computed from the current residual replicates. Thus the computational role of the MRTS layer is to reduce the dimension of the matrix decomposition from $n \times n$ to $K \times K$, rather than to introduce a large new stochastic parameter block.

5 Likelihood and Computation

5.1 Conditional Gaussian Likelihood

Given $(\boldsymbol{\beta}, \lambda, \mathbf{z}_t, \mathbf{D}_t, \mathbf{R})$, the conditional distribution of $\mathbf{Y}_t$ is

$$\mathbf{Y}_t \mid \boldsymbol{\beta}, \lambda, \mathbf{z}_t, \mathbf{D}_t, \mathbf{R} \sim N_n\big(\mathbf{X}_t\boldsymbol{\beta} + \lambda \mathbf{D}_t|\mathbf{z}_t|,\, \mathbf{D}_t\mathbf{R}\mathbf{D}_t\big).$$

Equivalently, the standardized residuals satisfy

$$\mathbf{r}_t \mid \mathbf{R} \sim N_n(\mathbf{0}, \mathbf{R}), \qquad t = 1, \ldots, T.$$

Hence, up to an additive constant, the conditional log-likelihood of the residual block is

$$\ell(\mathbf{R} \mid \mathbf{r}_1, \ldots, \mathbf{r}_T) = -\frac{T}{2}\log|\mathbf{R}| - \frac{1}{2}\sum_{t=1}^{T} \mathbf{r}_t'\mathbf{R}^{-1}\mathbf{r}_t. \tag{8}$$

5.2 Woodbury Identity for Fast Matrix Operations

Let

$$\mathbf{G} = \mathbf{F}\mathbf{M}\mathbf{F}' + \sigma_\xi^2\mathbf{I}_n, \qquad \sigma_\xi^2 > 0.$$

Then the Woodbury identity gives

$$\mathbf{G}^{-1} = \sigma_\xi^{-2}\mathbf{I}_n - \sigma_\xi^{-2}\mathbf{F}\left(\mathbf{M}^{-1} + \sigma_\xi^{-2}\mathbf{F}'\mathbf{F}\right)^{-1}\mathbf{F}'\sigma_\xi^{-2}. \tag{9}$$

Also, by the matrix determinant lemma,

$$\log|\mathbf{G}| = n\log(\sigma_\xi^2) + \log|\mathbf{M}| + \log\left|\mathbf{M}^{-1} + \sigma_\xi^{-2}\mathbf{F}'\mathbf{F}\right|. \tag{10}$$

Because

$$\mathbf{R} = \mathbf{A}^{-1/2}\mathbf{G}\mathbf{A}^{-1/2},$$

we obtain

$$\mathbf{R}^{-1} = \mathbf{A}^{1/2}\mathbf{G}^{-1}\mathbf{A}^{1/2}, \tag{11}$$

and

$$\log|\mathbf{R}| = \log|\mathbf{G}| - \sum_{i=1}^{n}\log(A_{ii}). \tag{12}$$

Thus all large-matrix computations needed in (8) can be reduced to operations involving $\mathbf{F}'\mathbf{F}$ and other $K \times K$ matrices, followed by matrix-vector products with $\mathbf{F}$.

5.3 Computational Complexity

For a single covariance update, the main costs are:

- forming $\mathbf{F}'\mathbf{r}_t$ for $t = 1, \ldots, T$, which costs $O(nKT)$;
- forming the $K \times K$ matrix $\mathbf{F}'\mathbf{S}\mathbf{F} = T^{-1}\sum_{t=1}^{T}(\mathbf{F}'\mathbf{r}_t)(\mathbf{F}'\mathbf{r}_t)'$, which costs $O(K^2T)$ after the projected residuals are available;
- eigendecomposition or inversion of $K \times K$ matrices, which costs $O(K^3)$.

This replaces repeated direct inversions of dense $n \times n$ Matérn covariance matrices, whose dominant cost is typically $O(n^3)$ per update. When $K \ll n$, this is the main source of computational acceleration.

6 Hybrid Bayesian–Profile Inference Scheme

The proposed method is not a fully Bayesian model for $\mathbf{M}$. Instead, it is a Bayesian hierarchical model for the skew, scale, regression, and partition blocks, combined with a profiled low-rank covariance update. A practical algorithm is:

1. Update latent censored or missing values in $\mathbf{Y}_t$ as in the original Morris computation.
2. Update $(\boldsymbol{\beta}, \lambda)$ using their conditional Gaussian blocks.
3. Update the partition-specific latent variables $\mathbf{z}_t$, the scale terms $\mathbf{D}_t$, and any partition-location parameters exactly as in the original Morris sampler.
4. Compute residuals $\mathbf{r}_t$ using (3).
5. Form $\mathbf{S} = T^{-1}\sum_{t=1}^{T}\mathbf{r}_t\mathbf{r}_t'$.
6. Update $(\mathbf{M}, \sigma_\xi^2)$ by the closed-form formulas (4)–(5).
7. Construct $\mathbf{G}$, $\mathbf{A}$, and $\mathbf{R}$ via (6)–(7).
8. Evaluate quadratic forms and log-determinants using (9)–(12).

This scheme is best viewed as a hybrid MCMC/ECM algorithm: the Morris blocks are updated by Bayesian conditional simulation, whereas the low-rank covariance block is updated by a closed-form optimization step conditional on the current residuals.

7 Optional One-Parameter Bayesian Extension

If some uncertainty in the covariance block is desired without returning to a full $K(K+1)/2$-dimensional MCMC update, one simple extension is to introduce a scalar expansion parameter $c > 0$ and set

$$\mathbf{M} = c\,\widehat{\mathbf{M}},$$

where $\widehat{\mathbf{M}}$ is the Tzeng–Huang plug-in estimator. One may then place a prior on c, for example

$$c \sim \mathrm{Gamma}(a_c, b_c),$$

and update only c within the MCMC. This preserves the dominant eigenstructure estimated from the closed-form step while allowing limited posterior uncertainty in the overall strength of the low-rank dependence.

8 Modeling Assumptions and Practical Remarks

1. The MRTS basis matrix $\mathbf{F}$ is treated as fixed and prespecified.
2. The Tzeng–Huang update relies on replicate residual vectors $\mathbf{r}_1, \ldots, \mathbf{r}_T$ that are conditionally independent with a common spatial dependence structure.
3. In the present proposal, σ_ϵ^2 is set to zero because the standardized residual vectors are treated as directly observed for the purpose of the covariance update. If an external measurement-error variance is known, it can be inserted into the update formulas.
4. Standardizing $\mathbf{G}$ to $\mathbf{R}$ is a deliberate modeling choice made to preserve the original Morris interpretation that $\mathbf{D}_t$ controls marginal scale while the spatial layer controls dependence.
5. The basis dimension K should remain well below n, and in practice it is also advisable to avoid choosing K much larger than the effective number of temporal replicates T.

9 Summary

The proposed model keeps the Morris skew-t hierarchy intact at the observation level,

$$\mathbf{Y}_t = \mathbf{X}_t\boldsymbol{\beta} + \lambda\mathbf{D}_t|\mathbf{z}_t| + \mathbf{D}_t\mathbf{v}_t,$$

but replaces the site-level Matérn dependence with an MRTS low-rank spatial layer. The unknown covariance matrix $\mathbf{M}$ is not sampled as a large Bayesian parameter block. Instead, it is updated in closed form from current residuals using the Tzeng–Huang estimator, after which the implied covariance matrix is standardized to a correlation matrix and inserted back into the Morris hierarchy. This retains computational scalability while preserving the original interpretation of the skew and scale blocks.